

E. Data Structures Fan

time limit per test: 2 seconds
memory limit per test: 256 megabytes

You are given an array of integers $a_1, a_2, \dots, a_n$, as well as a binary string[†] s consisting of n characters.

Augustin is a big fan of data structures. Therefore, he asked you to implement a data structure that can answer q queries. There are two types of queries:

- "1 $l\ r$ " ($1 \leq l \leq r \leq n$) — replace each character s_i for $l \leq i \leq r$ with its opposite. That is, replace all 0 with 1 and all 1 with 0.
- "2 g " ($g \in \{0, 1\}$) — calculate the value of the bitwise XOR of the numbers a_i for all indices i such that $s_i = g$. Note that the XOR of an empty set of numbers is considered to be equal to 0.

Please help Augustin to answer all the queries!

For example, if $n = 4$, $a = [1, 2, 3, 6]$, $s = 1001$, consider the following series of queries:

- "2 0" — we are interested in the indices i for which $s_i = 0$, since $s = 1001$, these are the indices 2 and 3, so the answer to the query will be $a_2 \oplus a_3 = 2 \oplus 3 = 1$.
- "1 1 3" — we need to replace the characters s_1, s_2, s_3 with their opposites, so before the query $s = 1001$, and after the query: $s = 0111$.
- "2 1" — we are interested in the indices i for which $s_i = 1$, since $s = 0111$, these are the indices 2, 3, and 4, so the answer to the query will be $a_2 \oplus a_3 \oplus a_4 = 2 \oplus 3 \oplus 6 = 7$.
- "1 2 4" — $s = 0111 \rightarrow s = 0000$.
- "2 1" — $s = 0000$, there are no indices with $s_i = 1$, so since the XOR of an empty set of numbers is considered to be equal to 0, the answer to this query is 0.

[†] A binary string is a string containing only characters 0 or 1.

Input

The first line of the input contains one integer t ($1 \leq t \leq 10^4$) — the number of test cases in the test.

The descriptions of the test cases follow.

The first line of each test case description contains an integer n ($1 \leq n \leq 10^5$) — the length of the array.

The second line of the test case contains n integers $a_1, a_2, \dots, a_n$ ($1 \leq a_i \leq 10^9$).

The third line of the test case contains the binary string s of length n .

The fourth line of the test case contains one integer q ($1 \leq q \leq 10^5$) — the number of queries.

The subsequent q lines of the test case describe the queries. The first number of each query, $tp \in \{1, 2\}$, characterizes the type of the query: if $tp = 1$, then 2 integers $1 \leq l \leq r \leq n$ follow, meaning that the operation of type 1 should be performed with parameters l, r , and if $tp = 2$, then one integer $g \in \{0, 1\}$ follows, meaning that the operation of type 2 should be performed with parameter g .

It is guaranteed that the sum of n over all test cases does not exceed 10^5 , and also that the sum of q over all test cases does not exceed 10^5 .

Output

For each test case, and for each query of type 2 in it, output the answer to the corresponding query.

Example

inputCopy

```
5
5
1 2 3 4 5
01000
7
2 0
2 1
1 2 4
2 0
2 1
1 1 3
2 1
6
12 12 14 14 5 5
001001
3
2 1
1 2 4
2 1
4
7 7 7 777
1111
3
2 0
1 2 3
2 0
2
1000000000 996179179
11
1
2 1
5
1 42 20 47 7
00011
5
1 3 4
1 1 1
1 3 4
1 2 4
2 0
```

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Practice



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binary searchbitmasksddata structuresdp*1500

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output

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3 2 6 7 7
11 7
0 0
16430827
47

Note

Let's analyze the first test case:

- 1. "2 0" — we are interested in the indices i for which $s_i = 0$, since $s = 01000$, these are the indices 1, 3, 4, and 5, so the answer to the query will be $a_1 \oplus a_3 \oplus a_4 \oplus a_5 = 1 \oplus 3 \oplus 4 \oplus 5 = 3$.
- 2. "2 1" — we are interested in the indices i for which $s_i = 1$, since $s = 01000$, the only suitable index is 2, so the answer to the query will be $a_2 = 2$.
- 3. "1 2 4" — we need to replace the characters s_2, s_3, s_4 with their opposites, so before the query $s = 01000$, and after the query: $s = 00110$.
- 4. "2 0" — we are interested in the indices i for which $s_i = 0$, since $s = 00110$, these are the indices 1, 2, and 5, so the answer to the query will be $a_1 \oplus a_2 \oplus a_5 = 1 \oplus 2 \oplus 5 = 6$.
- 5. "2 1" — we are interested in the indices i for which $s_i = 1$, since $s = 00110$, these are the indices 3 and 4, so the answer to the query will be $a_3 \oplus a_4 = 3 \oplus 4 = 7$.
- 6. "1 1 3" — $s = 00110 \rightarrow s = 11010$.
- 7. "2 1" — we are interested in the indices i for which $s_i = 1$, since $s = 11010$, these are the indices 1, 2, and 4, so the answer to the query will be $a_1 \oplus a_2 \oplus a_4 = 1 \oplus 2 \oplus 4 = 7$.

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